

Project Title
E-Commerce Product Entity-Resolution & Matching Platform
Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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ABSTRACT

The E-Commerce Product Entity-Resolution & Matching Platform is a smart solution designed to identify and match identical products listed by different online sellers, even when the product titles, descriptions, and specifications are different. Many e-commerce websites contain duplicate listings for the same product, making it difficult for customers to compare prices and choose the best option. This project collects product data from multiple marketplaces, cleans and standardizes the information, and applies entity-resolution and matching techniques to group similar products into a single canonical product catalog. The platform compares products using attributes such as brand, model, category, specifications, and features to accurately identify matching items. It also eliminates duplicate records, improves search accuracy, and enables reliable price comparison across different sellers. The system benefits both customers and businesses by providing consistent product information, reducing data redundancy, and improving inventory management. Overall, the platform enhances the online shopping experience by making product discovery faster, comparisons more accurate, and purchasing decisions easier while ensuring a well-organized and efficient e-commerce product database.

Signature of the Guide